# GUJARAT TECHNOLOGICAL UNIVERSITY

**Program Name: Diploma Engineering**

**Level: Diploma**

**Branch: Plastics Engineering**

**Subject Code: DI05023011**

**Subject Name: Plastics Testing and Quality Management**

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|---|---|
| **w. e. f. Academic Year:** | 2026-27 |
| **Semester:** | 5th |
| **Category of the Course:** | PCC |

| | |
|---|---|
| **Prerequisite:** | NA |
| **Rationale:** | Plastics being widely used in diversified applications, it is necessary to test the material properties before deciding to use them for each application. By testing of plastics, the product design and the area of application can be explored. Diploma Plastic engineer has to deals with the production of various types of plastics as a responsible technician and first line supervisor in the industries. Hence, the course has been design to develop these competencies and its associated cognitive, practical and effective domain learning out comes. This course also aims to create awareness about quality control and quality management methods. |

**Course Outcome:**

After Completion of the Course, Student will able to:

| No | Course Outcomes | RBT Level |
|---|---|---|
| 01 | Identify factors affecting the tests. | R,U,A |
| 02 | Perform various mechanical tests. | R,U,A |
| 03 | Interpret thermal properties. | R,U,A |
| 04 | Measure electrical and optical properties. | R,U,A |
| 05 | Observe rheological and weather tests. | R,U,A |

**Revised Bloom's Taxonomy (RBT)*

**Teaching and Examination Scheme:**

| Teaching Scheme (in Hours) | | | Total Credits L+T+ (PR/2) | Assessment Pattern and Marks | | | | Total Marks |
|---|---|---|---|---|---|---|---|---|
| | | | | Theory | | Tutorial / Practical | | |
| L | T | PR | C | ESE (E) | PA(M) | PA(I) | ESE (V) | |
| 3 | 0 | 2 | 4 | 70 | 30 | 20 | 30 | 150 |

**Course Content:**

| Unit No. | Content | No. of Hours | % of Weightage |
|---|---|---|---|
| **Unit – I**<br><br>**Introduction** | 1.1 Definition<br>1.2 Reasons for testing<br>1.3 Purpose of standard & specification<br>1.4 various organization dealing with standards and quality<br>1.5 Conditioning of samples | 04 | 14 %<br>(10 Marks) |
| **Unit – II**<br><br>**Mechanical Properties** | 2.1 Tensile strength test<br>2.2 Stress-Strain curve<br>2.3 Creep & stress relaxation<br>2.4 Flexural strength test<br>2.5 Impact strength test<br>• Izod impact<br>• Dart impact<br>2.6 Hardness<br>• Rockwell hardness<br>• Durometer hardness<br>2.7Abrasion resistance | 15 | 32 %<br>(22 Marks) |
| **Unit –III**<br><br>**Thermal Properties** | 3.1 Determination of Heat Deflection Temperature (HDT)<br>3.2 Determination of Vicat Softening Temperature (VST)<br>3.3 Thermal expansion test | 04 | 14 %<br>(10 Marks) |
| **Unit – IV**<br><br>**Electrical and Optical Properties** | 4.1 Die electric strength<br>4.2 Die electric constant<br>4.3 Arc resistance<br>4.4 Definition of volume resistance and surface resistance<br>4.5 Refractive index<br>4.6 Definition of haze, gloss, light transmission, clarity | 11 | 20 %<br>(14 Marks) |
| **Unit – V**<br><br>**Miscellaneous Tests** | 5.1 Melt flow index<br>5.2 Cup flow test and Spiral flow test<br>5.3 Environmental Stress Cracking Resistance<br>5.4 Oxygen index test<br>5.5 Specific gravity test<br>5.6 Outdoor and accelerated weather resistance tests.<br>5.7 Non-destructive test -Ultrasonic testing | 11 | 20 %<br>(14 Marks) |
| | **Total** | **45** | **100 %** |

| | | | |
|---|---|---|---|
| | | | (70 Marks) |

**Suggested Specification Table with Marks (Theory):**

| Distribution of Theory Marks (in %) | | | | | |
|---|---|---|---|---|---|
| **R Level** | **U Level** | **A Level** | **N Level** | **E Level** | **C Level** |
| **32** | **48** | **20** | **---------** | **_____** | **_____** |

*Where R: Remember; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create (as per Revised Bloom's Taxonomy)*

**References/Suggested Learning Resources:**
**(a) Books:**

| Sr. No. | Title of Book | Author | Publication with place, year and ISBN |
|---|---|---|---|
| 1 | Hand Book of Plastics Testing Technology | Vishu Shah | Wiley Inter-science publication |
| 2 | Hand Book of Polymer testing | R.P.Brown (roger brown) | Marcel- Dekker Inc. |
| 3 | Statistical Quality Control | O.P.Khanna | Khanna publishers |
| 4 | Hand book of plastics & elastomers | C.A.Harper | Wiley publication |
| 5 | Plastics processing data hand book | D.V.Rosato | Springer Berlin Heidelberg |

**(b) Open-source software and website:**

1. http://www.ipolytech.com/
2. http://www.ulttc.com/
3. www.intertek.com
4. http://www.labtesting.com/
5. www.nslanalytical.com/testing/polymer
6. http://www.exova.com/capabilities/polymer-testing/
7. http://www.chemir.com/plastic-polymer-testing.html

**Suggested Practical List:**

| Sr. No. | Practical/Exercise (Course Outcomes in Psychomotor Domain according to NBA Terminology) | Approx. Hrs Required |
|---|---|---|
| 1 | Measure the Tensile strength of a given plastic. | 2 |
| 2 | Measure the Flexural strength of a given plastic. | 2 |
| 3 | Measure the Izod impact strength of a given plastic. | 2 |
| 4 | Measure the Hardness of a given plastic. | 4 |
| 5 | Measure the HDT of a given plastic. | 4 |
| 6 | Measure the VST of a given plastic. | 4 |
| 7 | Measure the Dielectric strength of a given plastic. | 2 |
| 8 | Measure the ESCR of a given plastic. | 4 |
| 9 | Measure the Melt Flow Index of a given plastic. | 2 |
| 10 | Measure the Oxygen Index of a given plastic | 2 |
| 11 | Measure the Specific Gravity. | 2 |
| | Total | 30 |

**Suggested Activities for Students:**

1. Draw tensile test specimen and collect tensile specimen sample
2. List MFI for various plastic materials and decide their processing grade
3. Draw and collect izod impact strength test specimen sample and give importance of impact strength
4. Give oxygen index of various material and correlate it with their flammability

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